

Jiayin Pan

CONTACT INFORMATION	Department of Mathematics University of California, Santa Cruz 1156 High Street Santa Cruz, CA 95064	jpan53@ucsc.edu jypan10@gmail.com https://jypan10.github.io/
RESEARCH INTERESTS	Riemannian geometry, Ricci curvature and topology, Gromov-Hausdorff convergence.	
EMPLOYMENT	Fall 2022 - Present <i>Assistant Professor</i> Department of Mathematics, University of California, Santa Cruz, CA, USA. 2021 - 2022 Academic Year <i>Fields Postdoctoral Fellow</i> Fields Institute, Toronto, ON, Canada. 2018 - 2021 Academic Years <i>Visiting Assistant Professor</i> Department of Mathematics, University of California, Santa Barbara, CA, USA.	
EDUCATION	2012 - 2018 Academic Years <i>Ph.D. in Mathematics</i> Rutgers University, New Brunswick, NJ, USA. Advisor: Xiaochun Rong 2008 - 2012 Academic Years <i>B.S. in Mathematics</i> Shanghai Jiao Tong University, China.	
RESEARCH ARTICLES	<i>The articles are listed in the reverse order they were finished and submitted to arXiv.</i> Nonnegative Ricci curvature and virtual abelianness in dimensions less than 12 (with Dimitri Navarro and Xingyu Zhu) arXiv:2606.27724 Complete manifolds with nonnegative Ricci curvature and slow relative volume growth (with Dimitri Navarro and Xingyu Zhu) arXiv:2604.14537 Universal non-CD of sub-Riemannian manifolds (with Dimitri Navarro) <i>J. Reine Angew. Math.</i> 833 (2026) 111-136 On the topology of manifolds with nonnegative Ricci curvature and linear volume growth (with Dimitri Navarro and Xingyu Zhu) to appear in <i>J. Eur. Math. Soc.</i> , arXiv:2410.15488 Nonnegative Ricci curvature, splitting at infinity, and first Betti number rigidity (with Zhu Ye) <i>Adv. Math.</i> 459 (2024), 110028 Ricci curvature and the fundamental groups of effective regular sets <i>J. Math. Study</i> 58 (2025), no. 1, 3-21 (special issue in honor of Xiaochun Rong on his 70th birthday) Nonnegative Ricci curvature, nilpotency, and Hausdorff dimension <i>Math. Ann.</i> 394:2 (2026) The Grushin hemisphere as a Ricci limit space with curvature ≥ 1 <i>Proc. Amer. Math. Soc. Ser. B.</i> 10 (2023), 71-75 Singular Weyl's law with Ricci curvature bounded below (with Xianzhe Dai, Shouhei Honda, and Guofang Wei) <i>Trans. Amer. Math. Soc. Ser. B.</i> 10 (2023), 1212-1253	

Examples of open manifolds with positive Ricci curvature and non-proper Busemann functions (with Guofang Wei)
Amer. J. Math. 148 (2026), 235-246

Nonnegative Ricci curvature, metric cones, and virtual abelianness
Geom. & Topol. 28 (2024), 1409-1436

Examples of Ricci limit spaces with non-integer Hausdorff dimension (with Guofang Wei)
Geom. Funct. Anal. 32 (2022), 676-685

Some topological results of Ricci limit spaces (with Jikang Wang)
Trans. Amer. Math. Soc. 375 (2022), 8445-8464

Nonnegative Ricci curvature and escape rate gap
J. Reine Angew. Math. 782 (2022), 175-196

On the escape rate of geodesic loops in an open manifold with nonnegative Ricci curvature
Geom. & Topol. 25 (2021), 1059-1085

Semi-local simple connectedness of non-collapsing Ricci limit spaces (with Guofang Wei)
J. Eur. Math. Soc. 24 (2022), no. 12, 4027-4062

Nonnegative Ricci curvature, almost stability at infinity, and structure of fundamental groups
Adv. Math. 474 (2025), 110310

Ricci curvature and isometric actions with scaling nonvanishing property (with Xiaochun Rong)
Unpublished preprint, arXiv:1808.02329

Nonnegative Ricci curvature, stability at infinity and finite generation of fundamental groups
Geom. & Topol. 23 (2019), 3203-3231

A proof of Milnor conjecture in dimension 3
J. Reine Angew. Math. 758 (2020), 253-260

SURVEY ARTICLES

The fundamental groups of open manifolds with nonnegative Ricci curvature
SIGMA 16 (2020), 078, 16 pages (Special Issue on Scalar and Ricci Curvature in honor of Misha Gromov on his 75th Birthday)

Universal covers of Ricci limit and RCD spaces (with Guofang Wei)
Differential Geometry in the Large, London Mathematical Society Lecture Note Series (463), Cambridge University Press, 2020, Page 352-372. ISBN: 9781108884136.

Spaces with Ricci Curvature Lower Bounds (with Guofang Wei)
Surveys in Geometric Analysis 2023, pp 103-119, Science Press, Beijing, 2024. ISBN: 9787030789655.

PHD STUDENTS

Abhinav Jha, current
 Karl Puttler, current

POSTDOCS SUPERVISED

Dimitri Navarro, 23-26 AYs
 Fabio Ricci, current

FELLOWSHIPS & GRANTS

National Science Foundation, DMS-2304698, 9/1/2023 to 8/31/2026

Simons Foundation, Travel Support for Mathematicians, 9/1/2023 to 8/31/2028
Fields Postdoctoral Fellowship, 7/1/2021 to 6/30/2022
AMS-Simons Travel Grant, 7/1/2019 to 6/30/2021

ACADEMIC
ACTIVITIES

Referee for the following journals:

Annals of Mathematics
Geometric and Functional Analysis
Journal of the European Mathematical Society
Communications on Pure and Applied Mathematics
Geometry & Topology
American Journal of Mathematics
Journal of Differential Geometry
Journal für die Reine und Angewandte Mathematik
Advances in Mathematics
Calculus of Variations and Partial Differential Equations
Journal of Functional Analysis
Transactions of the American Mathematical Society
International Mathematics Research Notices
Proceedings of the American Mathematical Society
Bulletin of the London Mathematical Society
Journal of Geometric Analysis
Differential Geometry and its Application
International Journal of Mathematics
Symmetry, Integrability and Geometry: Methods and Applications
Journal of Topology and Analysis
Journal of Mathematical Study

Research Membership for SLMATH program, *New Frontiers in Curvature: Flows, General Relativity, Minimal Submanifolds, and Symmetry* (August-September 2024)

Organizer of UCSC Mathematics Colloquium (24-25 AY)

Organizer of UCSC Geometry & Analysis Seminar (22-23 AY, 26 Winter, 26-27 AY)

Co-organizer of UCSB Differential Geometry Seminar (19-21 AYs)